

QGenie

AI Question Paper Analysis System

Powered by Generative AI & NLP

Team

Auralis

Team Members

Mehrin Fathima

Ryan Jigo

Muhammad Abdulla

Athul VK

Project Report

1. Project Overview

QGenie is a Generative AI-powered application that analyzes question papers and provides insights such as difficulty level, topic distribution, cognitive skill assessment, and question categorization. The system helps educators evaluate exam quality and identify areas for improvement.

2. Technologies Used

Category	Technologies
Frontend	HTML5, CSS3, JavaScript, Streamlit
Backend	Python
AI & NLP	Large Language Models (LLMs), Generative AI, NLP
Libraries & Frameworks	LangChain, OpenAI/Groq API, PyPDF2/pdfplumber, Pandas, NumPy
Database & Storage	SQLite (Optional), Local File Storage
Development Tools	VS Code, Git & GitHub, Hugging Face Spaces

3. Features Included

- Question Paper Upload**
Upload question papers in PDF format with automatic text extraction from documents.
- AI-Based Analysis**
Analyze question patterns using Generative AI to understand the context of questions.
- Difficulty Classification**
Categorize questions into Easy, Medium, and Hard levels.
- Topic Identification**
Detect major topics covered in the examination and generate topic-wise distribution reports.
- Bloom's Taxonomy Analysis**
Classify questions into cognitive levels: Remember, Understand, Apply, Analyze, Evaluate, and Create.
- Marks Distribution Analysis**
Analyze marks allocation across topics and question types.
- Summary Generation**
Generate a concise report of the entire question paper.

■ Question Categorization

Identify MCQs, Short Answer Questions, Long Answer Questions, and Problem-Solving Questions.

■ Interactive Dashboard

User-friendly interface for viewing analysis results with visual charts and statistics.

■ Downloadable Report

Generate and download AI-generated analysis reports.

4. Advantages

- ✓ Saves teachers' time in manual evaluation.
- ✓ Provides objective analysis of question papers.
- ✓ Improves exam quality and balance.
- ✓ Supports outcome-based education.
- ✓ Generates instant insights using AI.

5. Future Enhancements

- Support for multiple languages.
- Comparison of multiple question papers.
- Semester-wise trend analysis.
- Automated exam quality scoring.
- Integration with Learning Management Systems (LMS).

6. Conclusion

QGenie leverages Generative AI and Natural Language Processing to automate question paper analysis. The system provides educators with valuable insights regarding difficulty level, topic coverage, cognitive skill assessment, and overall examination quality, making the evaluation process faster and more efficient.